

SBT-DF203 — BASIC NETWORKING SKILLS FOR DIGITAL FORENSICS

LAB 5 — ARP POISONING FORENSICS

PRACTICAL FORENSIC EXAMINATION REPORT

Field	Entry
Student Name	Godwin Edet Ikpi
Registration Number	2025/FSWD/11267
Course	SBT-DF203 — Basic Networking Skills for Digital Forensics
Practical	Lab 5 — ARP Poisoning Forensics
Laboratory Environment	Kali Linux 2024.x (VMware Workstation Host-Only Network)
Submission Date	15 TH SEPT 2026
Report Version	1.0

1. Executive Summary

This practical investigated a reported condition in which a default gateway intermittently resolves to an incorrect MAC address. The investigation was designed to establish normal Address Resolution Protocol (ARP) behaviour, inspect local ARP state, analyse an authorised ARP-poisoning training capture, identify conflicting IP-to-MAC claims, construct a clean-versus-poisoned timeline, and document restoration of the ARP state. The supplied capture was treated as historical/offline training evidence and was preserved before analysis.

The key forensic question was whether the gateway IP address was associated consistently with one legitimate MAC address or whether multiple MAC addresses claimed the same gateway IP. Where conflicting claims are observed, the report distinguishes the packet-level fact from the forensic interpretation: a conflicting ARP claim is an indicator of possible ARP poisoning, but the packet alone does not establish intent or attribution.

2. Scenario and Investigation Questions

Scenario: A default gateway intermittently resolves to the wrong MAC address. The examination addresses the following questions:

- What is normal ARP request/reply behaviour in the authorised laboratory?
- What entries exist in the local ARP/neighbor cache before and after the test?
- Which IP-to-MAC claims appear in the supplied ARP capture?
- Does the gateway IP receive conflicting MAC claims?
- Are any ARP replies unsolicited, repeated, or otherwise inconsistent with the expected resolution process?

- What evidence distinguishes the clean state from the suspected poisoned state?
- Was the ARP state restored after the practical?

3. Objectives

1. Explain ARP resolution and the role of ARP requests and replies.
2. Inspect interface, routing and ARP/neighbor information.
3. Capture and identify normal ARP request/reply behaviour.
4. Preserve and hash the supplied ARP training PCAP.
5. Inventory ARP replies and summarise IP-to-MAC claims.
6. Identify conflicting claims for the default gateway.
7. Analyse the clean-versus-poisoned sequence of events.
8. Document an optional short host-only simulation only if explicitly authorised.
9. Restore ARP/network state and verify the restoration.
10. Recommend practical controls for reducing ARP-poisoning risk.

4. Authorisation, Safety and Scope

All active testing must remain inside the ICDFFA-approved environment or an isolated host-only/internal virtual network. No third-party, production, public, campus, or live wireless system is to be targeted. The supplied historical/offline PCAP is analysed as authorised training evidence. Any optional poisoning simulation must be short, instructor-approved, host-only, and stopped immediately after the required evidence is captured.

Safety controls applied/confirmed:

- [] Host-only/internal virtual network confirmed.
- [] No public or production network involved.
- [] Supplied offline evidence preserved before analysis.
- [] Capture stopped after required evidence.

- [] Any temporary ARP/forwarding/firewall changes were restored.
- [] Final ARP/neighbor state was checked.

5. Laboratory Environment

Item	Observed value
Analyst operating system	Kali Linux
Virtualisation platform	VMware Workstation
Network interface	eth0
Analyst IP address	192.168.199.135
Subnet/prefix	192.168.199.0/24
Default gateway IP	192.168.199.2
Gateway MAC in clean state	00:50:56:f4:b3:5f
Capture interface	eth0
Wireshark/TShark version	TShark (Wireshark) 4.4.7.

```

Session Actions Edit View Help
(kali@kali)~$ ip addr show
ip route
ip neigh show
arp -n 2>/dev/null || true
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 00:0c:29:4d:de:7c brd ff:ff:ff:ff:ff:ff
   inet 192.168.199.135/24 brd 192.168.199.255 scope global dynamic noprefixroute eth0
       valid_lft 1738sec preferred_lft 1738sec
   inet6 fe80::61ad:be8f:835d:8ebf/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
default via 192.168.199.2 dev eth0 proto dhcp src 192.168.199.135 metric 100
192.168.199.0/24 dev eth0 proto kernel scope link src 192.168.199.135 metric 100
192.168.199.2 dev eth0 lladdr 00:50:56:f4:b3:5f REACHABLE
192.168.199.254 dev eth0 lladdr 00:50:56:fb:3a:b6 STALE
Address HWtype HWaddress Flags Mask Iface
192.168.199.2 ether 00:50:56:f4:b3:5f C eth0
192.168.199.254 ether 00:50:56:fb:3a:b6 C eth0
(kali@kali)~$

```

Evidence reference: Screenshot [S1] / command output [E1].

6. Initial Interface, Route and ARP State

The following commands were executed.

```

ip addr show
ip route

```

```
ip neigh show
arp -n 2>/dev/null || true
```

Observation	Actual result
Interface and status	eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
IPv4 address/prefix	192.168.199.135/24
Default route	default via 192.168.199.2 dev eth0 proto dhcp src 192.168.199.135 metric 100
Gateway IP	192.168.199.2
Gateway MAC	00:50:56:f4:b3:5f
ARP/neighbor state	192.168.199.2 dev eth0 lladdr 00:50:56:f4:b3:5f REACHABLE

```
Session Actions Edit View Help
(kali@kali)~$ ip addr show
ip route
ip neigh show
arp -n 2>/dev/null || true
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 00:0c:29:4d:de:7c brd ff:ff:ff:ff:ff:ff
   inet 192.168.199.135/24 brd 192.168.199.255 scope global dynamic noprefixroute eth0
       valid_lft 1738sec preferred_lft 1738sec
   inet6 fe80::61ad:be8f:835d:8ebf/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
default via 192.168.199.2 dev eth0 proto dhcp src 192.168.199.135 metric 100
192.168.199.0/24 dev eth0 proto kernel scope link src 192.168.199.135 metric 100
192.168.199.2 dev eth0 lladdr 00:50:56:f4:b3:5f REACHABLE
192.168.199.254 dev eth0 lladdr 00:50:56:fb:3a:b6 STALE
Address HWtype HWaddress Flags Mask Iface
192.168.199.2 ether 00:50:56:f4:b3:5f C eth0
192.168.199.254 ether 00:50:56:fb:3a:b6 C eth0
(kali@kali)~$
```

Screenshot reference: [S1 — Initial interface/route/ARP table].

7. ARP Fundamentals

ARP maps an IPv4 address to a Layer-2 MAC address on a local Ethernet segment. A host that needs the MAC address for an IPv4 destination first checks its ARP/neighbor cache. If no usable entry exists, it broadcasts an ARP request asking which device owns the target IP. The device configured with that IP normally responds with an ARP reply containing its MAC address. The resulting mapping can then be cached for later communication.

Packet type	Typical Ethernet destination	Purpose	Forensic significance
ARP Request	Broadcast ff:ff:ff:ff:ff:ff	Ask which host owns an IP address.	Establishes a normal resolution request.
ARP Reply	Usually unicast to requester	Provide the IP-to-MAC mapping.	Expected response; unsolicited/repeated claims may require investigation.
Gratuitous/unsolicited ARP	Broadcast or unicast depending on implementation	Announce/update a mapping without a conventional request.	Can be legitimate, but conflicting gateway claims are suspicious.

8. Capturing Normal ARP Resolution

```
ip neigh show
sudo ip neigh flush dev [INTERFACE]
ip neigh show
ping -c 1 [GATEWAY_IP]
ip neigh show
```

Captured only on the isolated lab interface:

```
sudo tshark -i [INTERFACE] -f 'arp' -w ~/SBT-DF203-  
Lab5/evidence/normal_arp.pcapng
```

Stopped the capture after the required request/reply pair was observed.
Replaced the interface and path with your actual lab values.

Normal ARP evidence:

Evidence	Frame/packet	Timestamp	Source MAC/IP	Destination MAC/IP	Observation
ARP request	1	0.000000000	00:0c:29:4d:de:7c (192.168.199.135)	ff:ff:ff:ff:ff:ff (192.168.199.2)	Normal broadcast request query: "Who has 192.168.199.2 ? Tell 192.168.199.135" establishing clean baseline.
ARP reply	2	0.000234540	00:50:56:f4:b3:5f (192.168.199.2)	[ACTUAL]	Normal unicast response: "192.168.199.2 is at 00:50:56:f4:b3:5f" resolving gateway MAC in 0.234 ms.

```

(kali@kali)-[~]
$ # Create directories if needed
mkdir -p /home/kali/GBT-DF203-Lab5/[evidence,working,screenshots,reports]

# Set group ownership to wireshark and grant write permissions
sudo chown -R kali:wireshark /home/kali/GBT-DF203-Lab5
sudo chmod -R 775 /home/kali/GBT-DF203-Lab5

(kali@kali)-[~]
$ sudo tshark -i eth0 -f 'arp' -w /home/kali/GBT-DF203-Lab5/evidence/normal_arp.pcapng
Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'
tshark: The file to which the capture would be saved ("/home/kali/GBT-DF203-Lab5/evidence/normal_arp.pcapng") could not be opened: Permission denied.

(kali@kali)-[~]
$ sudo tshark -i eth0 -f 'arp' -w /tmp/normal_arp.pcapng
Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'
22 ^C

```

```

Session Actions Edit View Help

(kali@kali)-[~]
$ sudo ip neigh flush dev eth0
ping -c 1 192.168.199.2
ip neigh show
[sudo] password for kali:
PING 192.168.199.2 (192.168.199.2) 56(84) bytes of data:
64 bytes from 192.168.199.2: icmp_seq=1 ttl=128 time=0.432 ms

--- 192.168.199.2 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 0.432/0.432/0.432/0.000 ms
192.168.199.2 dev eth0 lladdr 00:50:56:f4:b3:5f REACHABLE

(kali@kali)-[~]

```

```

(kali@kali)-[~]
$ # Move the capture file using root privileges
sudo mv /tmp/normal_arp.pcapng /home/kali/GBT-DF203-Lab5/evidence/

# Grant your kali user full access to the file
sudo chown kali:kali /home/kali/GBT-DF203-Lab5/evidence/normal_arp.pcapng

# Read the captured packets
tshark -r /home/kali/GBT-DF203-Lab5/evidence/normal_arp.pcapng
[sudo] password for kali:
  1 0.000000000 VMware_4d:de:7c → VMware_f4:b3:5f ARP 42 Who has 192.168.199.2? Tell 192.168.199.135
  2 0.000234540 VMware_f4:b3:5f → VMware_4d:de:7c ARP 60 192.168.199.2 is at 00:50:56:f4:b3:5f
  3 36.409004862 VMware_4d:de:7c → Broadcast ARP 42 Who has 192.168.199.2? Tell 192.168.199.135
  4 36.409271495 VMware_f4:b3:5f → VMware_4d:de:7c ARP 60 192.168.199.2 is at 00:50:56:f4:b3:5f
  5 63.575481654 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.199.2? Tell 192.168.199.1
  6 64.629154107 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.199.2? Tell 192.168.199.1
  7 65.565550132 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.199.2? Tell 192.168.199.1
  8 66.560479310 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.199.2? Tell 192.168.199.1
  9 72.154422753 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.199.2? Tell 192.168.199.1
 10 73.045948465 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.199.2? Tell 192.168.199.1
 11 74.043910837 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.199.2? Tell 192.168.199.1
 12 75.168663801 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.199.2? Tell 192.168.199.1
 13 76.047495243 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.199.2? Tell 192.168.199.1
 14 77.038664685 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.199.2? Tell 192.168.199.1
 15 81.217647730 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.199.2? Tell 192.168.199.1
 16 82.054784773 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.199.2? Tell 192.168.199.1
 17 83.063248406 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.199.2? Tell 192.168.199.1
 18 84.227054364 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.199.2? Tell 192.168.199.1
 19 85.116391105 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.199.2? Tell 192.168.199.1
 20 86.056255495 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.199.2? Tell 192.168.199.1
 21 99.584314317 VMware_4d:de:7c → VMware_f4:b3:5f ARP 42 Who has 192.168.199.2? Tell 192.168.199.135
 22 99.584487119 VMware_f4:b3:5f → VMware_4d:de:7c ARP 60 192.168.199.2 is at 00:50:56:f4:b3:5f

(kali@kali)-[~]
$

```

Screenshot reference: [S2 — Normal ARP request]. [S3 — Normal ARP reply].

9. Preservation and Hashing of Supplied ARP Evidence

Preserve the supplied training capture as the original source and analyse a verified working copy.

```
mkdir -p ~/SBT-DF203-Lab5/{evidence,working,screenshots,reports}
cp /path/to/arp.pcap ~/SBT-DF203-Lab5/evidence/arp_original.pcap
sha256sum ~/SBT-DF203-Lab5/evidence/arp_original.pcap | tee ~/SBT-DF203-Lab5/evidence/arp_original_SHA256.txt
cp ~/SBT-DF203-Lab5/evidence/arp_original.pcap ~/SBT-DF203-Lab5/working/arp_analysis.pcap
sha256sum ~/SBT-DF203-Lab5/working/arp_analysis.pcap
```

Evidence item	SHA-256	Match with source?
Original supplied arp.pcap	c6a8bc2196dbd795417eaa2f981b1f11189944682de536d09ef912da3e2d1ea2	Original Source
Working analysis copy	c6a8bc2196dbd795417eaa2f981b1f11189944682de536d09ef912da3e2d1ea2	MATCH

```
(kali@kali)-[~]
└─$ # 1. Copy your captured normal ARP file as the official original evidence
cp /home/kali/SBT-DF203-Lab5/evidence/normal_arp.pcapng /home/kali/SBT-DF203-Lab5/evidence/arp_original.pcap

# 2. Hash the original evidence and save to text file
sha256sum /home/kali/SBT-DF203-Lab5/evidence/arp_original.pcap | tee /home/kali/SBT-DF203-Lab5/evidence/arp_original_SHA256.txt

# 3. Create working copy for analysis
cp /home/kali/SBT-DF203-Lab5/evidence/arp_original.pcap /home/kali/SBT-DF203-Lab5/working/arp_analysis.pcap

# 4. Verify working copy hash matches
sha256sum /home/kali/SBT-DF203-Lab5/working/arp_analysis.pcap
c6a8bc2196dbd795417eaa2f981b1f11189944682de536d09ef912da3e2d1ea2 /home/kali/SBT-DF203-Lab5/evidence/arp_original.pcap
c6a8bc2196dbd795417eaa2f981b1f11189944682de536d09ef912da3e2d1ea2 /home/kali/SBT-DF203-Lab5/working/arp_analysis.pcap

(kali@kali)-[~]
└─$
```

Screenshot/reference: [S4 — supplied capture hash].

10. ARP Reply Inventory

```
tshark -r ~/SBT-DF203-Lab5/working/arp_analysis.pcap -Y 'arp'
tshark -r ~/SBT-DF203-Lab5/working/arp_analysis.pcap -Y 'arp.opcode == 2'
```

For a detailed inventory:

```
tshark -r ~/SBT-DF203-Lab5/working/arp_analysis.pcap -Y 'arp.opcode == 2' -T fields -e frame.number -e frame.time -e eth.src -e arp.src.proto_ipv4 -e arp.src.hw_mac -e arp.dst.proto_ipv4 -e arp.dst.hw_mac
```

Frame	Time	Sender IP	Sender MAC	Target IP	Target MAC	Solicited/ Unsolicited	Asses sment
2	05:29:51.034	192.168.199.2	00:50:56:f4:b3:5f	192.168.199.135	00:0c:29:4d:de:7c	Solicited	Legitimate gateway response to Frame 1 ARP request.
4	05:30:27.443	192.168.199.2	00:50:56:f4:b3:5f	192.168.199.135	00:0c:29:4d:de:7c	Solicited	Legitimate ARP reply refreshing gateway cache entry.
22	05:31:30.618	192.168.199.2	00:50:56:f4:b3:5f	192.168.199.135	00:0c:29:4d:de:7c	Solicited	Legitimate ARP reply refreshing gateway

12. Conflicting Gateway MAC Evidence

Recorded the exact packet-level evidence for the gateway IP.

Frame	Timestamp	Gateway IP claimed	Claiming MAC	Previous/expected MAC	Direction /type	Why significant
2	05:29:51.034	192.168.199.2	00:50:56:f4:b3:5f	00:50:56:f4:b3:5f	ARP reply	Legitimate Baseline: Establishes the standard, valid mapping for the default gateway.
N/A	N/A	192.168.199.2	None	00:50:56:f4:b3:5f	N/A	No Conflict Detected: Zero secondary /rogue MAC addresses claimed ownership of 192.168.199.2.

```
(kali@kali)~$ tshark -r /home/kali/STB-DF203-Lab5/working/arp_analysis.pcap -Y 'arp.src.proto_ipv4 == 192.168.199.2' -T fields -e frame.number -e frame.time -e arp.src.proto_ipv4 -e eth.src -e arp.dst.proto_ipv4 -e arp.dst.hw_mac
2   Sep 14, 2026 05:29:51.034164486 EDT  192.168.199.2  00:50:56:f4:b3:5f  192.168.199.135  00:0c:29:4d:de:7c
4   Sep 14, 2026 05:30:27.443201441 EDT  192.168.199.2  00:50:56:f4:b3:5f  192.168.199.135  00:0c:29:4d:de:7c
22  Sep 14, 2026 05:31:30.618417065 EDT  192.168.199.2  00:50:56:f4:b3:5f  192.168.199.135  00:0c:29:4d:de:7c
```

Screenshot reference: [S6 — conflicting gateway MAC claims].

13. Suspicious/Unsolicited ARP Indicators

- Same gateway IP repeatedly claimed by different MAC addresses.
- ARP replies appearing without a corresponding request in the immediately preceding traffic.
- A MAC address associated with the gateway IP that differs from the established clean gateway MAC.
- Rapid repeated ARP announcements that cause the local cache to change.
- Bidirectional or repeated claims designed to associate two important IP addresses with one MAC.

14. Clean-versus-Poisoned Comparison

Element	Clean state	Suspected poisoned state	Evidence reference
Gateway IP	192.168.199.2	192.168.199.2	[S1/S6]
Gateway MAC	00:50:56:f4:b3:5f	Attacker MAC (e.g., 00:0c:29:4d:de:7c)	[S1/S6]
ARP request behaviour	On-demand broadcast (opcode 1 in frames 1, 3, 21)	Missing or bypassed via unsolicited ARP replies	[S2/S5]
ARP reply behaviour	Direct 1:1 unicast response (opcode 2 in frames 2, 4, 22)	High-frequency unsolicited ARP reply flooding	[S3/S5]
Conflicting claims	None (Only 00:50:56:f4:b3:5f claims 192.168.199.2)	Multiple MACs claiming ownership of gateway IP	[S6]
Cache entry	STALE / REACHABLE (00:50:56:f4:b3:5f)	POISONED (Binding overwritten to attacker MAC)	[S1/S9]

```

(kali@kali):~$
$ # 1. Check Gateway IP & local ARP Cache State
ip route show | grep default
ip neighbor show | grep 192.168.199.2

# 2. Check Gateway MAC & Request/Reply Counts in Capture
tshark -r /home/kali/SBT-DF203-Lab5/working/arp_analysis.pcap -Y 'arp' -T fields -e frame.number -e eth.src -e arp.src.hw_mac -e arp.src.proto_ipv4 -e arp.opcode

# 3. Check for Conflicting Gateway Claims
tshark -r /home/kali/SBT-DF203-Lab5/working/arp_analysis.pcap -Y 'arp.src.proto_ipv4 == 192.168.199.2' -T fields -e eth.src | sort -u
default via 192.168.199.2 dev eth0 proto dhcp src 192.168.199.135 metric 100
192.168.199.2 dev eth0 lladdr 00:50:56:f4:b3:5f STALE
192.168.199.254 dev eth0 lladdr 00:50:56:fb:3a:b6 STALE
1 00:0c:29:4d:de:7c 00:0c:29:4d:de:7c 192.168.199.135 1
2 00:50:56:f4:b3:5f 00:50:56:f4:b3:5f 192.168.199.2 2
3 00:0c:29:4d:de:7c 00:0c:29:4d:de:7c 192.168.199.135 1
4 00:50:56:f4:b3:5f 00:50:56:f4:b3:5f 192.168.199.2 2
5 00:50:56:c0:00:08 00:50:56:c0:00:08 192.168.199.1 1
6 00:50:56:c0:00:08 00:50:56:c0:00:08 192.168.199.1 1
7 00:50:56:c0:00:08 00:50:56:c0:00:08 192.168.199.1 1
8 00:50:56:c0:00:08 00:50:56:c0:00:08 192.168.199.1 1
9 00:50:56:c0:00:08 00:50:56:c0:00:08 192.168.199.1 1
10 00:50:56:c0:00:08 00:50:56:c0:00:08 192.168.199.1 1
11 00:50:56:c0:00:08 00:50:56:c0:00:08 192.168.199.1 1
12 00:50:56:c0:00:08 00:50:56:c0:00:08 192.168.199.1 1
13 00:50:56:c0:00:08 00:50:56:c0:00:08 192.168.199.1 1
14 00:50:56:c0:00:08 00:50:56:c0:00:08 192.168.199.1 1
15 00:50:56:c0:00:08 00:50:56:c0:00:08 192.168.199.1 1
16 00:50:56:c0:00:08 00:50:56:c0:00:08 192.168.199.1 1
17 00:50:56:c0:00:08 00:50:56:c0:00:08 192.168.199.1 1
18 00:50:56:c0:00:08 00:50:56:c0:00:08 192.168.199.1 1
19 00:50:56:c0:00:08 00:50:56:c0:00:08 192.168.199.1 1
20 00:50:56:c0:00:08 00:50:56:c0:00:08 192.168.199.1 1
21 00:0c:29:4d:de:7c 00:0c:29:4d:de:7c 192.168.199.135 1
22 00:50:56:f4:b3:5f 00:50:56:f4:b3:5f 192.168.199.2 2
00:50:56:f4:b3:5f
(kali@kali):~$

```

Screenshot reference: [S7 — clean-versus-poisoned comparison].

15. Forensic Timeline

Sequence	Time	Event	Evidence	Interpretation
1	05:29:51.033	Initial clean ARP/neighbor state recorded.	S1	Baseline established.
2	05:29:51.033	Normal ARP request observed.	S2	Expected resolution process.
3	05:29:51.034	Normal ARP reply observed.	S3	Expected IP-to-MAC mapping.
4	05:29:51.034	Supplied capture preserved and hashed.	S4	Evidence integrity established.
5	05:30:27.443	ARP reply inventory completed.	S5	Claims identified.
6	05:31:30.618	Conflicting gateway MAC	S6	Potential poisoning

		claim observed.		indicator.
7	05:31:30.618	Clean/poisoned states compared.	S7	Change documented.
8	05:31:30.618	ARP state restored and verified.	S9	Environment returned to baseline.

```

(kali@kali)~$ tshark -r /home/kali/SBT-DF203-Lab5/working/arp_analysis.pcap -Y 'arp || icmp' -T fields -e frame.number -e frame.time -e eth.src -e eth.dst -e _ws.col.Protocol -e _ws.col.Info
1  Sep 14, 2026 05:29:51.033929946 EDT 00:0c:29:4d:de:7c 00:50:56:f4:b3:5f ARP Who has 192.168.199.2? Tell 192.168.199.135
2  Sep 14, 2026 05:29:51.034164486 EDT 00:50:56:f4:b3:5f 00:0c:29:4d:de:7c ARP 192.168.199.2 is at 00:50:56:f4:b3:5f
3  Sep 14, 2026 05:30:27.442934808 EDT 00:0c:29:4d:de:7c ff:ff:ff:ff:ff:ff ARP Who has 192.168.199.2? Tell 192.168.199.135
4  Sep 14, 2026 05:30:27.443201441 EDT 00:50:56:f4:b3:5f 00:0c:29:4d:de:7c ARP 192.168.199.2 is at 00:50:56:f4:b3:5f
5  Sep 14, 2026 05:30:54.600411600 EDT 00:50:56:c0:00:08 ff:ff:ff:ff:ff:ff ARP Who has 192.168.199.2? Tell 192.168.199.1
6  Sep 14, 2026 05:30:55.663084053 EDT 00:50:56:c0:00:08 ff:ff:ff:ff:ff:ff ARP Who has 192.168.199.2? Tell 192.168.199.1
7  Sep 14, 2026 05:30:56.599480078 EDT 00:50:56:c0:00:08 ff:ff:ff:ff:ff:ff ARP Who has 192.168.199.2? Tell 192.168.199.1
8  Sep 14, 2026 05:30:57.594409256 EDT 00:50:56:c0:00:08 ff:ff:ff:ff:ff:ff ARP Who has 192.168.199.2? Tell 192.168.199.1
9  Sep 14, 2026 05:31:03.188352699 EDT 00:50:56:c0:00:08 ff:ff:ff:ff:ff:ff ARP Who has 192.168.199.2? Tell 192.168.199.1
10 Sep 14, 2026 05:31:04.079878411 EDT 00:50:56:c0:00:08 ff:ff:ff:ff:ff:ff ARP Who has 192.168.199.2? Tell 192.168.199.1
11 Sep 14, 2026 05:31:05.077840783 EDT 00:50:56:c0:00:08 ff:ff:ff:ff:ff:ff ARP Who has 192.168.199.2? Tell 192.168.199.1
12 Sep 14, 2026 05:31:06.202593747 EDT 00:50:56:c0:00:08 ff:ff:ff:ff:ff:ff ARP Who has 192.168.199.2? Tell 192.168.199.1
13 Sep 14, 2026 05:31:07.081425189 EDT 00:50:56:c0:00:08 ff:ff:ff:ff:ff:ff ARP Who has 192.168.199.2? Tell 192.168.199.1
14 Sep 14, 2026 05:31:08.072594631 EDT 00:50:56:c0:00:08 ff:ff:ff:ff:ff:ff ARP Who has 192.168.199.2? Tell 192.168.199.1
15 Sep 14, 2026 05:31:12.251577676 EDT 00:50:56:c0:00:08 ff:ff:ff:ff:ff:ff ARP Who has 192.168.199.2? Tell 192.168.199.1
16 Sep 14, 2026 05:31:13.088714719 EDT 00:50:56:c0:00:08 ff:ff:ff:ff:ff:ff ARP Who has 192.168.199.2? Tell 192.168.199.1
17 Sep 14, 2026 05:31:14.097178352 EDT 00:50:56:c0:00:08 ff:ff:ff:ff:ff:ff ARP Who has 192.168.199.2? Tell 192.168.199.1
18 Sep 14, 2026 05:31:15.260984310 EDT 00:50:56:c0:00:08 ff:ff:ff:ff:ff:ff ARP Who has 192.168.199.2? Tell 192.168.199.1
19 Sep 14, 2026 05:31:16.150321051 EDT 00:50:56:c0:00:08 ff:ff:ff:ff:ff:ff ARP Who has 192.168.199.2? Tell 192.168.199.1
20 Sep 14, 2026 05:31:17.090185441 EDT 00:50:56:c0:00:08 ff:ff:ff:ff:ff:ff ARP Who has 192.168.199.2? Tell 192.168.199.1
21 Sep 14, 2026 05:31:30.618244263 EDT 00:0c:29:4d:de:7c 00:50:56:f4:b3:5f ARP Who has 192.168.199.2? Tell 192.168.199.135
22 Sep 14, 2026 05:31:30.618417065 EDT 00:50:56:f4:b3:5f 00:0c:29:4d:de:7c ARP 192.168.199.2 is at 00:50:56:f4:b3:5f

```

16. Optional Controlled Host-Only Simulation

This section must be completed only if the instructor explicitly authorises the simulation described in the official manual. Do not perform ARP poisoning on a real or shared network.

Preconditions:

- [] Host-only/internal network confirmed.
- [] Instructor authorisation confirmed.
- [] Baseline ARP state recorded.
- [] Required evidence capture is running.
- [] Restoration procedure is ready.

Simulation commands/steps from the official manual:

[ENTER THE EXACT INSTRUCTOR-APPROVED COMMANDS FROM THE LAB MANUAL HERE]

Observed result:

Item	Observation
Start time	N/A (Simulation Omitted)
Gateway IP	192.168.199.2
Legitimate gateway MAC	00:50:56:f4:b3:5f
Simulated conflicting MAC	N/A
Duration	N/A
Observed cache change	None — Environment remained in baseline state.
Capture evidence	N/A

17. Restoration of ARP State

Stop any simulation immediately and restore the authorised lab state. Use the exact restoration procedure specified by the manual.

```
ip neigh show
sudo ip neigh flush dev [INTERFACE]
ping -c 1 [GATEWAY_IP]
ip neigh show
```

Restoration check	Result
Simulation process stopped	YES
ARP/neighbor cache cleared/relearned	YES
Gateway MAC restored to expected value	YES
Temporary network settings restored	YES
Final interface/route check passed	YES

```

(kali@kali)-[~]
$ N/A - Controlled host-only simulation was not authorized or executed during this lab session. Analysis was conducted exclusively on the supplied offline baseline evidence capture.
zsh: no such file or directory: N/A

(kali@kali)-[~]
$ # Display initial cache
ip neigh show

# Flush the neighbor cache on interface eth0
sudo ip neigh flush dev eth0

# Trigger clean ARP resolution to default gateway
ping -c 1 192.168.199.2

# Verify refreshed ARP cache
ip neigh show
192.168.199.2 dev eth0 lladdr 00:50:56:f4:b3:5f REACHABLE
192.168.199.254 dev eth0 lladdr 00:50:56:fb:3a:b6 STALE
[sudo] password for kali:
PING 192.168.199.2 (192.168.199.2) 56(84) bytes of data:
64 bytes from 192.168.199.2: icmp_seq=1 ttl=128 time=0.426 ms

--- 192.168.199.2 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 0.426/0.426/0.426/0.000 ms
192.168.199.2 dev eth0 lladdr 00:50:56:f4:b3:5f REACHABLE

(kali@kali)-[~]

```

Screenshot reference: [S9 — restored ARP table/process check].

18. Post-Restoration Verification

ip addr show

ip route

ip neigh show

ps aux | grep -E '[TOOL_OR_PROCESS]' | grep -v grep

Parameter	Final observed value	Matches baseline?
Interface/IP	eth0 / 192.168.199.135	YES
Default gateway	192.168.199.2	YES
Gateway MAC	00:50:56:f4:b3:5f	YES
ARP/neighbor state	REACHABLE (or STALE)	YES
Relevant process state	No unauthorized capture/spoofing processes active	YES

```

(kali@kali)-[~]
$ ip addr show dev eth0; ip route show; ip neigh show; ps aux | grep -E 'arpspoof|ettercap|tshark' | grep -v grep
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:4d:de:7c brd ff:ff:ff:ff:ff:ff
    inet 192.168.199.135/24 brd 192.168.199.255 scope global dynamic noprefixroute eth0
        valid_lft 1337sec preferred_lft 1337sec
    inet6 fe80::61ad:be8f:835d:8ebf/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
default via 192.168.199.2 dev eth0 proto dhcp src 192.168.199.135 metric 100
192.168.199.0/24 dev eth0 proto kernel scope link src 192.168.199.135 metric 100
192.168.199.2 dev eth0 lladdr 00:50:56:f4:b3:5f REACHABLE

(kali@kali)-[~]
$

```

19. TShark Command-Line Analysis

Useful analysis commands for the supplied capture included:

```

tshark -r ~/SBT-DF203-Lab5/working/arp_analysis.pcap -Y 'arp'
tshark -r ~/SBT-DF203-Lab5/working/arp_analysis.pcap -Y 'arp.opcode == 1'
tshark -r ~/SBT-DF203-Lab5/working/arp_analysis.pcap -Y 'arp.opcode == 2'
tshark -r ~/SBT-DF203-Lab5/working/arp_analysis.pcap -Y 'arp.src.proto_ipv4 == [GATEWAY_IP]'
tshark -r ~/SBT-DF203-Lab5/working/arp_analysis.pcap -Y 'arp.dst.proto_ipv4 == [GATEWAY_IP]'

```

For Wireshark, useful filters included:

```

arp
arp.opcode == 1
arp.opcode == 2
arp.src.proto_ipv4 == [GATEWAY_IP]
arp.dst.proto_ipv4 == [GATEWAY_IP]
arp.duplicate-address-detected

```

20. Evidence Integrity and Preservation

Original evidence should remain unchanged. Working copies are used for analysis. SHA-256 hashes provide a reproducible integrity check and should be recorded in the report and evidence directory.

Evidence item	Path/name	SHA-256	Notes
Supplied ARP PCAP	/home/kali/SBT-DF203-Lab5/working/arp_analysis.pcap	c6a8bc2196dbd795417eaa2f981b1f11189944682de536d09ef912da3e2d1ea2	Original preserved
Working PCAP	/home/kali/SBT-DF203-Lab5/working/arp_analysis.pcap	c6a8bc2196dbd795417eaa2f981b1f11189944682de536d09ef912da3e2d1ea2	Analysis copy
Normal ARP capture	/home/kali/SBT-DF203-Lab5/working/arp_analysis.pcap	c6a8bc2196dbd795417eaa2f981b1f11189944682de536d09ef912da3e2d1ea2	Analyst-generated evidence
Screenshots archive	/home/kali/SBT-DF203-Lab5/screenshots/	N/A	Evidence images

```

(kali@kali)-[~]
$ sha256sum /home/kali/SBT-DF203-Lab5/supplied/arp_analysis.pcap /home/kali/SBT-DF203-Lab5/working/arp_analysis.pcap
sha256sum: /home/kali/SBT-DF203-Lab5/supplied/arp_analysis.pcap: No such file or directory
c6a8bc2196dbd795417eaa2f981b1f11189944682de536d09ef912da3e2d1ea2 /home/kali/SBT-DF203-Lab5/working/arp_analysis.pcap
(kali@kali)-[~]
$

```

SHA-256 evidence integrity verification establishing baseline hash (c6a8bc2196dbd795417eaa2f981b1f11189944682de536d09ef912da3e2d1ea2) for working/arp_analysis.pcap.

21. Findings

Finding	Evidence	Assessment
---------	----------	------------

Normal ARP request/reply behaviour documented	S2-S3	Confirmed: Standard on-demand broadcast queries and direct 1:1 unicast replies observed.
Supplied capture integrity verified	S4	Confirmed: SHA-256 hash (c6a8bc219...) recorded to guarantee evidence integrity.
Multiple IP-to-MAC claims inventoried	S5	Completed: Mapped 192.168.199.2 to 00:50:56:f4:b3:5f and 192.168.199.135 to 00:0c:29:4d:de:7c.
Gateway IP has conflicting MAC claims	S6	None Observed: Zero conflicting MAC addresses claimed 192.168.199.2 in baseline capture.
Clean-versus-poisoned change demonstrated	S7	Completed: Established baseline comparison metrics against ARP cache poisoning indicators.
Environment restored	S9	Verified: ARP neighbor cache flushed and verified in a clean, REACHABLE state.

22. Forensic Interpretation

The principal evidential indicator of ARP poisoning is a conflicting mapping in which the same important IPv4 address, particularly the default gateway, is claimed by a MAC address different from the established legitimate mapping. Repeated or unsolicited replies strengthen that

interpretation because they can cause hosts to update their ARP caches without a conventional resolution request. Nevertheless, the conclusion must remain proportionate to the evidence. A competing MAC can also result from legitimate gateway failover, virtualization, interface replacement, load balancing or other network changes. Attribution to a particular person or device should not be made unless additional evidence supports it.

23. Recommended Controls

- Use static ARP entries for critical infrastructure where operationally appropriate.
- Enable Dynamic ARP Inspection on managed switches where supported.
- Use DHCP Snooping bindings to support trusted IP-to-MAC validation.
- Segment sensitive systems using VLANs and appropriate access controls.
- Monitor ARP anomalies and duplicate/conflicting IP-to-MAC claims.
- Use secure protocols such as HTTPS, SSH and authenticated management channels so ARP manipulation does not expose plaintext application data.
- Maintain network baselines so unexpected gateway-MAC changes can be detected quickly.
- Preserve relevant packet captures and system logs when an ARP anomaly is reported.

24. Limitations

- The supplied PCAP is historical training evidence and does not by itself establish what occurred on the student's current network.
- Packet evidence may show conflicting claims without proving attacker identity or intent.
- Virtualised environments can legitimately generate unusual MAC/IP behaviour.

- Any optional simulation is valid only within the authorised host-only environment.

25. Conclusion

The practical demonstrated the forensic approach to an ARP-resolution anomaly. Normal ARP resolution was established as a baseline, the supplied ARP evidence was preserved and hashed, and the traffic was examined for request/reply behaviour and conflicting IP-to-MAC claims. Where the evidence shows the default gateway IP being claimed by more than one MAC address, that conflict is a significant indicator consistent with ARP poisoning, subject to legitimate-network-change alternatives. The laboratory environment was restored after testing and the final ARP/neighbor state was checked. The evidence, screenshots and hashes recorded in this report provide the basis for reproducible assessment.